

Stage 3 Proposal: A Constructed Regime-Switching Task

Vanilla Policy versus Oracle-Signaled Policy on a Task We Control

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Stage 2 tests the chunk-length switch on natural long-horizon tasks, but it cannot control how much of each episode is stable transport and how much is unstable contact, and that mix is exactly what decides how much room the switch has to win. Stage 3 removes that limitation by constructing the task ourselves: a repeated transport-and-insert task in robosuite, with N pegs or sockets spaced far apart on a large table, so that each cycle is a long free-space transport (open-loop stable, wants a long chunk) followed by a tight insertion (open-loop unstable, wants per-step feedback), repeated N times. The two quantities the theory cares about become experimental knobs: the number of regime alternations N (swept over 1, 2, 4) and the ratio of stable to unstable segment length (set by peg spacing and insertion tolerance). The prediction is specific: the gap between the oracle-switched policy and the best fixed chunk length should grow with N , because every added alternation forces any fixed k into one more compromise. The comparison is vanilla policy at its best fixed k (found by a full sweep, same protocol as Stage 1a, never a strawman) against the same policy with oracle switching, and then with predictor switching. Because we define the task, the regime boundaries are known by construction, which gives three things the earlier stages had to pay heavily for: a perfect and free oracle signal (no perturbation labeling), a ground-truth check of the perturbation labeler (the measured λ bands should line up with the designed segment boundaries), and clean predictor training labels. Demonstrations come from MimicGen: author the task, record about ten source demonstrations, and generate one thousand automatically; the estimated cost is one to two days of task authoring plus a few GPU-days of generation and training per N . The known weakness is that a bespoke task invites the criticism that it was engineered for the method to win, so Stage 3 is positioned as the mechanism experiment that explains and quantifies the effect, while Stage 2 remains the naturalistic evidence; the pairing is stronger than either alone. Stage 3 starts after the Stage 2 results are in, since those results determine whether it quantifies how the win scales or diagnoses which regime mix the hypothesis needs.